Visit **www.brpaper.com** for downloading previous years question papers of
10th and 12th (PSEB and CBSE), IKPTU,MRSSTU, PSBTE, PANJAB UNIVERSITY,
PUNJABI UNIVERSITY, BFUHS, HPTU, HPSBTE , HARYANA DIPLOMA, MDU HARYANA

Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 09

B.Tech. (Electrical Engg.) (Sem.–7/8)

ELECTRICAL ENERGY CONSERVATION & AUDITING

Subject Code : BTEE-701A-18

M.Code : 90571

Date of Examination : 16-07-22

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

brpaper.com brpaper.com

SECTION-A

1. Write briefly :

a) Define commercial and non-commercial energy sources along with two examples of each energy source.

b) What are the causes of air pollution?

c) What do you understand by the term "maximizing energy efficiencies"?

d) Mention the methods for preparing process flow.

e) Name the factors which are responsible for losses in induction motors.

f) Why power factor improvement is significant in power system?

g) What are the components of compressed air system?

h) How performance evaluation of cooling towers is being done?

i) What are maximum demand controllers?

j) How an energy efficient transformer is different from conventional transformer?

BR paper
exams preparatory

SECTION-B

2. Write a brief note on :

   a) Energy sector reforms and

   b) Energy pricing.

3. What is meaning of restructuring of energy supply sector? Why this concept emerged and what is the need of restructuring of energy supply sector?

4. What is bench marking? What are its types? Why organizations adopt bench marking? Mention the various parameters which are used as bench mark in various organizations.

5. From point of view of power factor improvement, how the size as well as location of the capacitors is selected? How the performance assessment of power factor capacitors is being done?

6. Write a note on rewinding and motor replacement issues.

SECTION-C

7. What are the types of air compressors? What is an efficient compressor operation? How to evaluate compressor efficiency? How to assess capacity of air compressors?

8. Compare flow control strategies and energy conservation opportunities of :

   a) Fans and blowers

   b) Pumps

   c) Cooling tower

9. Write a note on :

   a) Variable speed drives

   b) Energy efficient lighting control.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

brpaper.com brpaper.com